

Lecture 02

Descriptive Statistics

Easy-to-understand exam notes based on the full 25-slide lecture

What this note gives you

Short explanations, all lecture points, formula reminders, slide examples, common exam traps, and memory tips - without unnecessary detail.

Graphical Methods

Histograms

Box Plots

Mean & Mode

Dispersion

Variance & SD

CV

Exam rule from the lecture: If a question does not say whether the data are from a population or sample, **treat them as a sample.**

Prepared from: IT2120 Lecture 02 - Descriptive Statistics, SLIIT Faculty of Computing, Year 02 Semester 01.

1. Descriptive Statistics - The Big Picture

Descriptive statistics is also called **preliminary analysis**. Its purpose is to give an initial idea about how the data and each variable behave.

Graphical methods

Use graphs to see the pattern, shape, frequencies, spread and possible unusual values.

Can be used for both categorical and numerical variables.

Numerical methods

Use numerical summaries to describe a numerical variable using one or more values.

Applied only to numerical variables.

 **Remember:** The method you choose depends on the **type of data** available.

Graph choice shown in the lecture

One Categorical Variable

- Bar charts
- Pie charts
- One-way frequency tables

One Numerical Variable

- Histograms
- Box plots
- Stem-and-leaf plots

 **Scope note:** Bar charts, pie charts, one-way frequency tables and stem-and-leaf plots are listed in this lecture, but they are not explained further in these slides.

 **Exam Important:** Be able to match the variable type with a suitable graph. A **histogram/box plot** is for numerical data; **bar/pie** is for categorical data.

2. Histograms

A histogram summarizes a numerical variable by splitting the data into **classes (intervals)** and showing how many observations fall inside each class.

- 1 **Divide the data into suitable classes.** The lecture uses classes with the **same width**.
- 2 **Find the frequency distribution.** A frequency distribution is the list of classes together with their frequencies (counts).
- 3 **Draw the histogram.** The x-axis shows classes; the y-axis can show frequency, relative frequency or percentage.
- 4 **Bars touch each other.** There are **no gaps** between neighboring histogram bars.

$$\text{Range} = \text{Maximum} - \text{Minimum}$$

$$\text{Class width} \approx \text{Range} \div \text{Number of classes}$$

Lecture example

For the marks data, **Max = 91** and **Min = 16**.

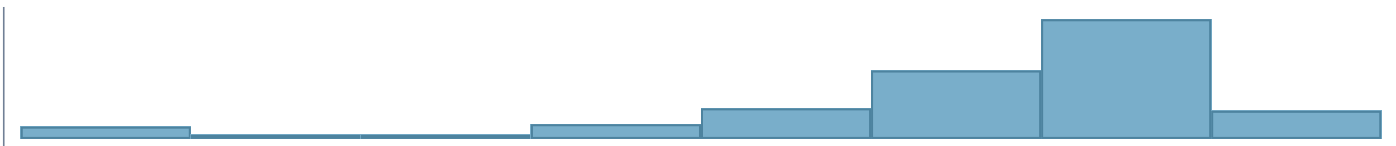
Range

$$91 - 16 = 75$$

Using 8 classes

$$75 \div 8 = 9.375 \approx 10$$

The lecture notes that a suitable number of classes is often **between 5 and 10**, and classes can also be selected by fixing the class width.



 **Memory tip: HISTO = Hinged bars** - imagine the bars are physically joined together, so no

gaps.

Slides 6-8

3. Histogram Example - Frequency Table

The lecture's histogram of the 20 marks uses these class boundaries and frequencies:

Class	Frequency	Meaning
14.5 - 24.5	1	One mark is in this interval.
24.5 - 34.5	0	No marks in this class.
34.5 - 44.5	0	No marks in this class.
44.5 - 54.5	1	One mark falls here.
54.5 - 64.5	2	Two observations.
64.5 - 74.5	5	Five observations.
74.5 - 84.5	9	Highest-frequency class.
84.5 - 94.5	2	Two observations.

★ **What the bar height means:** Each bar represents one class, and the bar height is proportional to the frequency of that class.

⚠ **Common exam confusion:** A histogram is not just a bar chart with numbers. Histogram bars represent **continuous numerical intervals** and are drawn adjacent to one another.

Quick exam method

1. Find Min and Max.
2. Calculate Range.
3. Choose number of classes (often 5-10 as stated in the lecture).
4. Find a practical class width.
5. Count observations in each class.
6. Draw touching bars.

4. Box Plots and Outliers

To draw a box plot, first identify the **five-number summary** and any **outliers**.

Minimum

Smallest value used in the summary.

Q1

First quartile.

Q2 / Median

Middle location.

Q3

Third quartile.

Maximum

Largest value used in the summary.

Lecture instruction

Use linear interpolation to find quartiles.

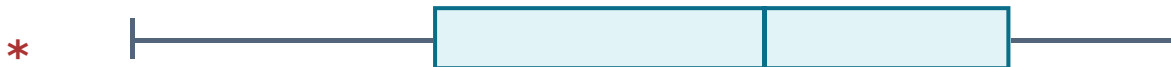
Outlier limits

$$\text{IQR} = Q3 - Q1$$

$$\text{Upper Bound} = Q3 + 1.5 \times \text{IQR}$$

$$\text{Lower Bound} = Q1 - 1.5 \times \text{IQR}$$

Values outside these limits are treated as **potential outliers** and are marked with an asterisk (*) in the box plot.



Sketch: outlier * | whisker | Q1 [box - median - box] Q3 | whisker

★ **Whisker rule:** The whiskers end at the minimum and maximum values that are **not outliers**.

🧠 **Memory tip:** Box plot = **5 numbers + 1.5 IQR rule**.

5. Box Plot Example from the Lecture

The lecture uses the same marks data for the box plot. In the shown box plot, **16 is marked as an outlier**.

What the plot is showing

- The central box is formed using **Q1, Median and Q3**.
- The whiskers extend to the smallest and largest values that are **not outliers**.
- The unusually low value **16** is plotted separately with an asterisk.

⚠ **Important source detail:** These slides say to use **linear interpolation** for quartiles, but they do not show the exact interpolation-position formula. In an exam, use the precise quartile method taught by your lecturer/lab.

Safe box-plot workflow for an exam

1. Sort the data.
2. Find Q1, Median (Q2), Q3 using the lecture's interpolation method.
3. Calculate $IQR = Q3 - Q1$.
4. Calculate lower and upper bounds.
5. Mark values outside the bounds as outliers.
6. Draw the box from Q1 to Q3, median inside, and whiskers only to non-outlier extremes.

★ **Exam trap:** Do not automatically draw whiskers to the raw minimum/maximum if either one is an outlier.

6. Numerical Methods & Central Tendency

Numerical methods are applied only to **numerical variables**. They summarize a variable using numerical values.

Four sections named in the lecture

1. Central tendency

Where the data are located as a whole.

2. Dispersion

How spread out the data are.

3. Skewness

Named in the lecture; no formula/details are developed in these slides.

4. Kurtosis

Named in the lecture; no formula/details are developed in these slides.

Measures of central tendency

These give an idea about the **location** of the data as a whole.

Measure	Simple meaning
Mean	Arithmetic average.
Median	Middle value / Q2 after ordering the data.
Mode	Value with the highest frequency.
Other location measures	Percentiles, deciles and quartiles.



Memory tip: Central tendency = **MMM**: Mean, Median, Mode.

Slides 13-15

7

7. Mean and Mode

Mean

The lecture names three types of means: **arithmetic**, **geometric** and **harmonic**. Only the **arithmetic mean** is discussed here, and it is simply called the mean.

Population mean

$$\mu = (\sum x_i) / N$$

N = number of population elements.

Sample mean

$$\bar{x} = (\sum x_i) / n$$

n = number of sample elements.

★ **Exam rule:** If the question does not specify population/sample, the lecture says **assume sample**.

Lecture example 1.2 - marks

There are 20 marks and their total is 1442.

$$\bar{x} = 1442 / 20 = 72.1$$

Lecture example 1.3 - frequency table

For the number of flaws, values repeat according to their frequencies. The total number of observations is 20 and the weighted total is 46.

$$\text{Mean} = 46 / 20 = 2.3 \text{ flaws}$$

Mode

- The mode is the value with the **highest frequency**.
- A data set can have **multiple modes**.

• If every value is different, the data set has **no mode**.

8. Measures of Dispersion

Dispersion describes the **spread** of the data as a whole.

Measure	Formula / meaning	Lecture note
Range	Max - Min	More suitable for small data sets.
IQR	Q3 - Q1	Not sensitive to outliers.
Variance	Average squared spread around the mean	Highly sensitive to outliers.
Standard deviation	$\sqrt{\text{Variance}}$	Measures spread on the original scale.

★ **Outlier sensitivity:** The lecture states that **range and variance are highly sensitive to outliers**, while **IQR is not sensitive to outliers**.

Population variance

$$\sigma^2 = \Sigma(x_i - \mu)^2 / N$$

Sample variance

$$s^2 = \Sigma(x_i - \bar{x})^2 / (n - 1)$$

Standard deviation

Population SD

$$\sigma = \sqrt{\sigma^2}$$

Sample SD

$$s = \sqrt{s^2}$$

 **Memory tip:** Sample = $n - 1$. Population = divide by **N**.

Slides 20-22

9. Relative Dispersion - Coefficient of Variation

The **Coefficient of Variation (CV)** is the lecture's relative measure of dispersion. It compares the magnitude of variation with the magnitude of the mean.

$$CV = (sd / \bar{x}) \times 100$$

How to read it: CV expresses the standard deviation as a percentage of the mean. It is useful when you want spread *relative* to the average size of the data.

Final lecture exercise - plastic hinges

The slide gives 25 observations and asks you to:

1. Find the five-number summary.
2. Find mode, mean, variance and standard deviation.
3. Calculate relative dispersion.
4. Draw the box plot.
5. Comment on the distribution.

Data: 72, 35, 63, 67, 87, 71, 64, 47, 60, 81, 39, 52, 57, 74, 43, 55, 37, 83, 48, 91, 53, 44, 94, 65, 75

Self-check values that do not depend on the quartile convention

Item	Calculated from the slide data
Mode	No mode (all 25 values are different).
Sample mean	$\bar{x} = 62.28$
Sample variance	$s^2 \approx 294.63$
Sample SD	$s \approx 17.16$
CV	$\approx 27.56\%$

Quartile note: Complete the five-number summary and box plot using the exact **linear interpolation** rule taught in your class, because this lecture slide does not state the position formula.

Slides 23-24

10. Exam Focus + Memory Sheet

Topic	What you should be able to do
Descriptive statistics	Explain preliminary analysis and the difference between graphical and numerical methods.
Graph choice	Match categorical data with bar/pie/frequency table; numerical data with histogram/box/stem-and-leaf.
Histogram	Find range, choose classes/class width, create frequencies, and remember no gaps .
Box plot	Find five-number summary, IQR, bounds, outliers, and whiskers to non-outlier extremes.
Mean	Use μ for population and $\bar{x}$ for sample; assume sample if unspecified.
Mode	Highest frequency; multiple modes possible; all unique = no mode.
Dispersion	Know Range, IQR, Variance and SD; know which are outlier-sensitive.
Variance	Population denominator N; sample denominator n - 1.
CV	Use $CV = sd / mean \times 100$.



Fast memory

MMM = Mean, Median, Mode

5 + 1.5 = five-number summary + 1.5 IQR outlier rule

n - 1 = sample variance

HISTO = **Hinged bars** = no gaps



Highest-priority exam details

Sample-vs-population formulas; histogram construction; outlier bounds; box-plot whisker rule; variance/SD; CV; outlier sensitivity.

One-page formula recap

Location

$$\bar{x} = \sum x_i / n$$

$$\mu = \sum x_i / N$$

Spread

$$\text{Range} = \text{Max} - \text{Min}$$

$$\text{IQR} = Q3 - Q1$$

Outliers

$$\text{LB} = Q1 - 1.5\text{IQR}$$

$$\text{UB} = Q3 + 1.5\text{IQR}$$

Variance

$$s^2 = \sum (x_i - \bar{x})^2 / (n-1)$$

$$\sigma^2 = \sum (x_i - \mu)^2 / N$$

SD

$$\text{SD} = \sqrt{\text{Variance}}$$

Relative spread

$$\text{CV} = (sd/\bar{x}) \times 100$$

The lecture ends after the final practice problem; skewness and kurtosis are named but not developed further in this slide set.
Full lecture recap